Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

CSE 200 – Technical Writing and Presentation

Diffusion Models

A Technical Overview of Progressive Denoising for Image Generation

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Expanded from the accompanying presentation into a formal technical report.

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Abstract

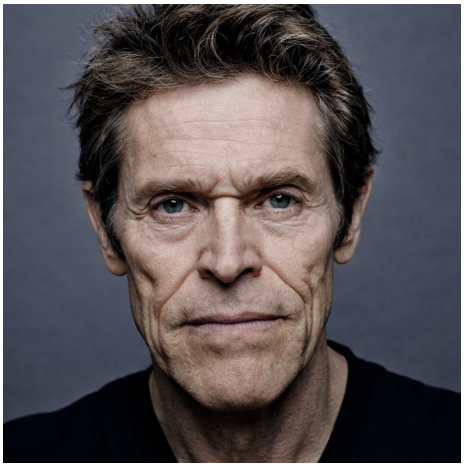
Diffusion models are generative models that learn to reverse a gradual noising process. Rather than producing an image in a single step, a diffusion model begins with random noise and refines it through a sequence of denoising operations until a coherent sample is obtained. This report develops the material from our presentation slides into a concise technical discussion of the main ideas behind Denoising Diffusion Probabilistic Models (DDPMs) [1, 2]. It outlines the forward and reverse processes, explains the role of the U-Net backbone and timestep conditioning, and presents the standard noise-prediction objective used in training. The report also discusses why diffusion models have become influential in modern generative modeling, particularly because of their stable optimization behavior, strong sample quality, and flexibility in conditional generation [3, 4]. Finally, it notes practical and ethical limitations, including slow sampling, high computational cost, and the need for responsible deployment.

1 Introduction

Generative modeling studies how to learn a probability distribution from observed data so that new samples can be drawn from it. In the case of images, the goal is not only to reproduce local texture, but also to preserve larger structure, object relationships, and overall realism. Diffusion-based methods have become especially important in this area because they combine a clear probabilistic interpretation with strong empirical performance [1, 5].

Figure 1 illustrates this motivation. The generated face on the right appears plausible even though it does not correspond to a real person. A useful generative model must therefore do more than memorize training examples; it must capture the statistical regularities of the data well enough to synthesize new samples.

(a) A real face sample.



(b) A model-generated face sample.

Figure 1: Illustrative comparison between a real image and a generated image. A successful generative model should produce outputs that remain visually consistent with the underlying data distribution.

The central idea of diffusion modeling is to decompose generation into many small denoising steps. During training, real data are gradually perturbed with Gaussian noise. Because the corruption process is known in advance, the model can be trained to predict the added noise at arbitrary timesteps. During sampling, the learned process is applied in reverse: the model starts from random noise and progressively recovers structure. This formulation turns image generation into a sequence of manageable estimation problems rather than a single difficult prediction.

2 Core Diffusion Mechanism

The workflow of a diffusion model has two complementary parts: a *forward diffusion process* that destroys information and a *reverse diffusion process* that reconstructs structure. Figure 2 summarizes this idea in one diagram.

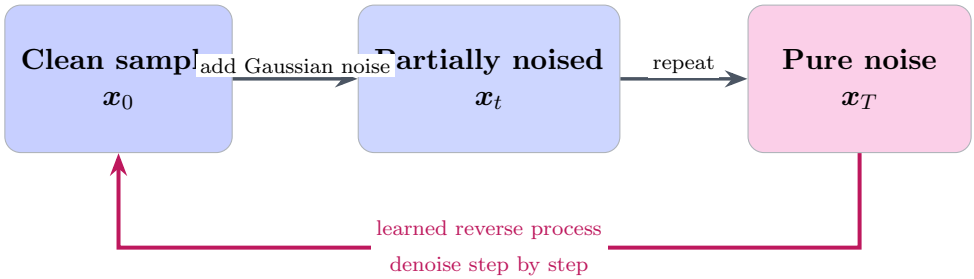


Figure 2: Conceptual overview of diffusion models. Training uses the controlled noising path, while sampling uses the learned reverse path.

2.1 Forward Diffusion

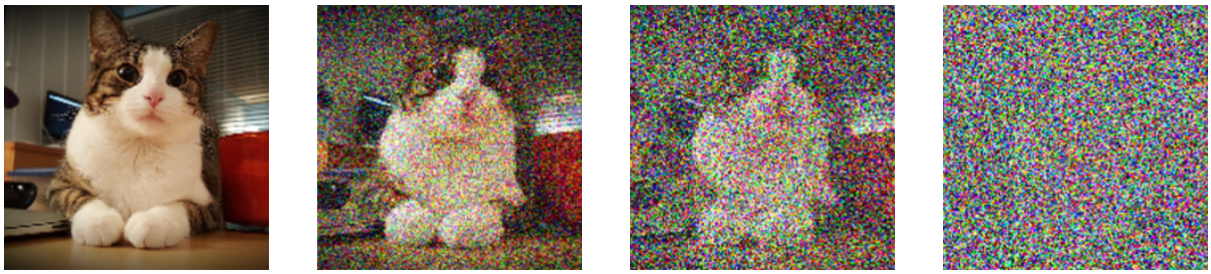
The forward process is a Markov chain that progressively perturbs a clean sample $\boldsymbol{x}_0 \in \mathbb{R}^d$ with Gaussian noise. At each timestep t, a small variance term β_t determines how much corruption is introduced:

$$q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\,\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I}\right). \tag{1}$$

If we define $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, then a sample at any timestep can be written directly as

$$\boldsymbol{x}_t = \sqrt{\bar{\alpha}_t}\,\boldsymbol{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}). \tag{2}$$

Equation 2 is convenient during training because it allows the model to sample a noisy state at any timestep directly, without simulating all previous steps.



(a) Clean (b) Low noise (c) Heavy noise (d) Pure noise

Figure 3: Visual interpretation of the forward diffusion process. A structured image is gradually transformed into an approximately Gaussian sample.

2.2 Reverse Diffusion

Once the data has been corrupted, the generative task becomes the reverse problem: infer how to move from $\boldsymbol{x}_t$ back to a less noisy sample $\boldsymbol{x}_{t-1}$. The model parameterized by θ learns a reverse distribution of the form

$$p_\theta(\boldsymbol{x}_{t-1} \mid \boldsymbol{x}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t), \boldsymbol{\Sigma}_\theta(\boldsymbol{x}_t, t)) \,. \tag{3}$$

In practice, many implementations train a neural network to predict the added noise rather than predict clean pixels directly. This reparameterization simplifies learning because the network only needs to estimate the perturbation that separates $\boldsymbol{x}_t$ from the underlying clean sample [1, 2].

3 Model Architecture and Training

In practice, strong performance depends not only on the probabilistic formulation, but also on the neural architecture used to model the reverse process. Most image diffusion systems employ a U-Net-style backbone with multi-scale feature extraction and skip connections [6]. The network receives two inputs: a noisy image $\boldsymbol{x}_t$ and an embedding of the timestep t.

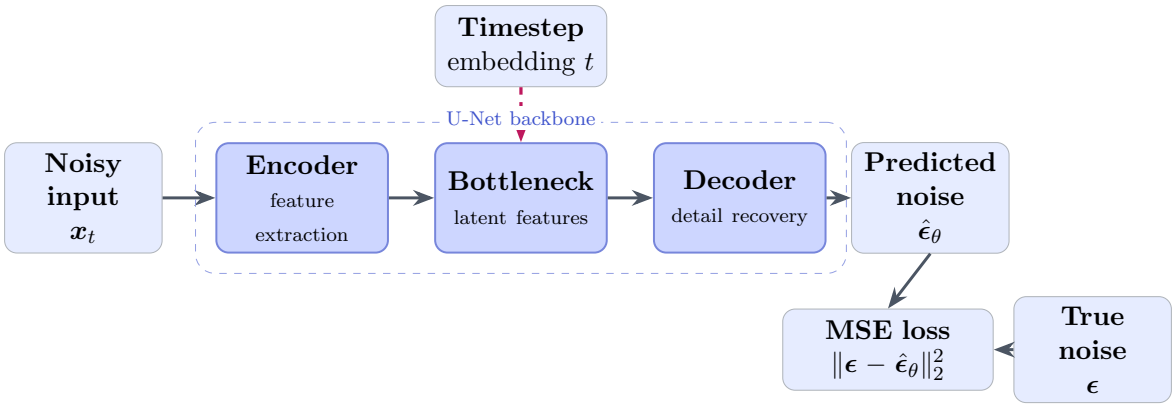


Figure 4: A simplified training view of an image diffusion model. The network receives a noisy image and the current timestep, predicts the noise component, and is trained with a mean-squared error objective.

3.1 Noise-Prediction Objective

The standard DDPM training objective is

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\boldsymbol{x}_0,t,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)\|_2^2 \right] . \tag{4}$$

This loss is convenient because the target noise $\boldsymbol{\epsilon}$ is known exactly; it is sampled by the training procedure itself. As a result, supervision is explicit and comparatively stable. Instead of learning to generate a final image in one pass, the network solves many smaller denoising tasks.

3.2 Training Workflow

The complete training routine can be summarized as Algorithm 1. The procedure alternates between sampling a real image, corrupting it to a randomly chosen timestep, and updating the network so that its predicted noise matches the actual perturbation.

Algorithm 1 Training a diffusion model with noise prediction

Require: Training dataset $\mathcal{D}$, noise schedule $\{\beta_t\}_{t=1}^T$, neural network $\boldsymbol{\epsilon}_\theta$

```
while training has not converged do
    Sample a clean image x_0 ~ D
    Sample a timestep t uniformly from {1, ..., T}
    Sample Gaussian noise ε ~ N(0, I)
    Form a noisy input using Equation 2
    Predict ε̂ = ε_θ(x_t, t)
    Compute L_simple = ||ε − ε̂||_2^2
    Update θ using gradient descent
end while
```

Table 1 collects the most important symbols used in the mathematical discussion so that the notation remains easy to follow.

Symbol	Meaning
$\boldsymbol{x}_0$	A clean training sample drawn from the data distribution.
$\boldsymbol{x}_t$	The noisy version of the sample at diffusion step t.
β_t	Variance schedule controlling how much noise is added at step t.
α_t	Shorthand for $1 - \beta_t$.
$\bar{\alpha}_t$	Cumulative product $\prod_{s=1}^{t} \alpha_s$, measuring the remaining signal after t steps.
$\boldsymbol{\epsilon}$	Gaussian noise used to corrupt the sample.
$\boldsymbol{\epsilon}_\theta(\boldsymbol{x}_t, t)$	Neural network prediction of the added noise.

Table 1: Notation used in the forward process, reverse process, and training objective.

4 Why Diffusion Models Perform Well

Diffusion models perform well in practice for several related reasons. First, the training target is available analytically once Gaussian noise has been sampled, which helps avoid the adversarial instability often associated with GAN-based training. Second, iterative denoising allows coarse structure to emerge before fine details are refined. Third, conditioning information such as text, class labels, or masks can be incorporated naturally into the denoising network, which makes the framework adaptable to many tasks.

These characteristics help explain why diffusion systems became competitive with, and in many settings preferable to, earlier generative approaches. Table 2 gives a concise comparison that highlights common trade-offs.

Model family	Training stability	Sample quality	Sampling speed	Main comment
VAE	High	Moderate	Fast	Strong probabilistic interpretation but often blurrier outputs.
GAN	Moderate to low	High	Very fast	Can generate sharp images but training may be unstable and prone to mode collapse.
Diffusion model	High	Very high	Slow	Excellent fidelity and diversity, but generation requires many denoising steps.

Table 2: A high-level comparison of common generative model families. The table emphasizes typical tendencies rather than strict universal rules.

5 Applications

Diffusion models now appear in both public-facing creative tools and specialized technical workflows. In consumer applications, they are used for text-to-image synthesis, image

editing, artistic rendering, and increasingly for video generation. Figure 5 shows a few widely recognized systems associated with this broader family of methods.

(a) DALL-E. (b) Midjourney. (c) Sora.

Figure 5: Examples of widely recognized systems associated with diffusion-based generation. The figure is illustrative rather than exhaustive.

Beyond creative media, diffusion models are increasingly relevant in technical domains. In medical imaging, researchers use them for denoising, reconstruction, and data augmentation. In scientific design, related models are being explored for molecule generation and inverse problems. Latent diffusion models further reduce the cost of high-resolution synthesis by performing denoising in a compressed latent space rather than directly in pixel space [4].

6 Limitations and Responsible Use

Despite their strengths, diffusion models still have important limitations. The first is **slow sampling**: high-quality generation may require dozens to hundreds of denoising steps, which increases latency. The second is **high computational cost**: training large systems demands substantial GPU memory, long training times, and significant electrical power. The third is **system complexity**: practical deployments often rely on careful prompt conditioning, safety filters, and post-processing pipelines.

There are also broader ethical concerns. Diffusion models can reproduce harmful biases present in their training data, imitate artistic styles without consent, or be used to create misleading synthetic media. For this reason, technical progress should be considered alongside dataset transparency, usage policies, and clear attribution practices.

7 Conclusion

Diffusion models offer a clear and influential framework for modern generative modeling. By learning to reverse a controlled corruption process, they connect probabilistic reasoning, neural network design, and iterative optimization in a way that is both mathematically grounded and practically effective. This combination explains their strong performance across image generation and related tasks.

At the same time, the topic becomes much easier to understand when its components are presented in a clear sequence: forward diffusion, reverse diffusion, architecture, training objective, applications, and limitations. A structured presentation of these ideas shows not only how diffusion models generate realistic samples, but also why they remain an important subject in contemporary machine learning.

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